

Project II Spacecraft Dynamics

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1 Euler's rotational equations of motion

Let $\mathbf{h}$ be the total angular momentum, $\mathbf{h}_s$ the angular momentum of the spacecraft and $\mathbf{h}_w$ the angular momentum of the reaction wheel system.

$$\mathbf{h} = \mathbf{h}_s + \mathbf{h}_w \quad (1.1)$$

Thus, in the body frame,

$${}^B\mathbf{h} = {}^B J_s^B \boldsymbol{\omega}_{B/N} + C_{BW}^W J_w C_{BW}^T C_{BW}^W \boldsymbol{\omega}_{W/N} \quad (1.2)$$

$${}^B\mathbf{h} = {}^B J_s^B \boldsymbol{\omega}_{B/N} + C_{BW}^W J_w ({}^W \boldsymbol{\omega}_{W/B} + C_{WB}^B \boldsymbol{\omega}_{B/N}) \quad (1.3)$$

Therefore

$${}^B\mathbf{h} = ({}^B J_s + {}^B J_w) \boldsymbol{\omega}_{B/N} + C_{BW}^W J_w^W \boldsymbol{\omega}_{W/B} \quad (1.4)$$

and finally, we get :

$${}^B\mathbf{h} = J\boldsymbol{\omega} + C_{BW}^W J_w \boldsymbol{\omega}_w \quad (1.5)$$

Using Euler's Equation

$$\dot{\mathbf{h}} = \mathbf{l} \quad (1.6)$$

where $\mathbf{l}$ is the external torque, assuming no external torque and using the fact that the wheel axes are aligned with the satellite body axes, we obtain :

$$J\dot{\boldsymbol{\omega}} = -[\boldsymbol{\omega}X]J\boldsymbol{\omega} - [\boldsymbol{\omega}X]J_w \boldsymbol{\omega}_w - J_w \dot{\boldsymbol{\omega}}_w \quad (1.7)$$

2 Tumbling motion

The reaction wheel aligned with $\hat{\mathbf{b}}_3$ is turned on. The wheel axis inertia is I_w and the speed of the wheel is Ω . Therefore, the Euler's rotational equation of motion becomes :

$$J\dot{\boldsymbol{\omega}} = -[\boldsymbol{\omega}X]J\boldsymbol{\omega} - [\boldsymbol{\omega}X]I_w \Omega_w \hat{\mathbf{b}}_3 - I_w \dot{\Omega}_w \hat{\mathbf{b}}_3 \quad (2.1)$$

3 Time responses

The equations of the previous parts are solved using MATLAB code and ODE45 function.

3.1 Non diagonal elements

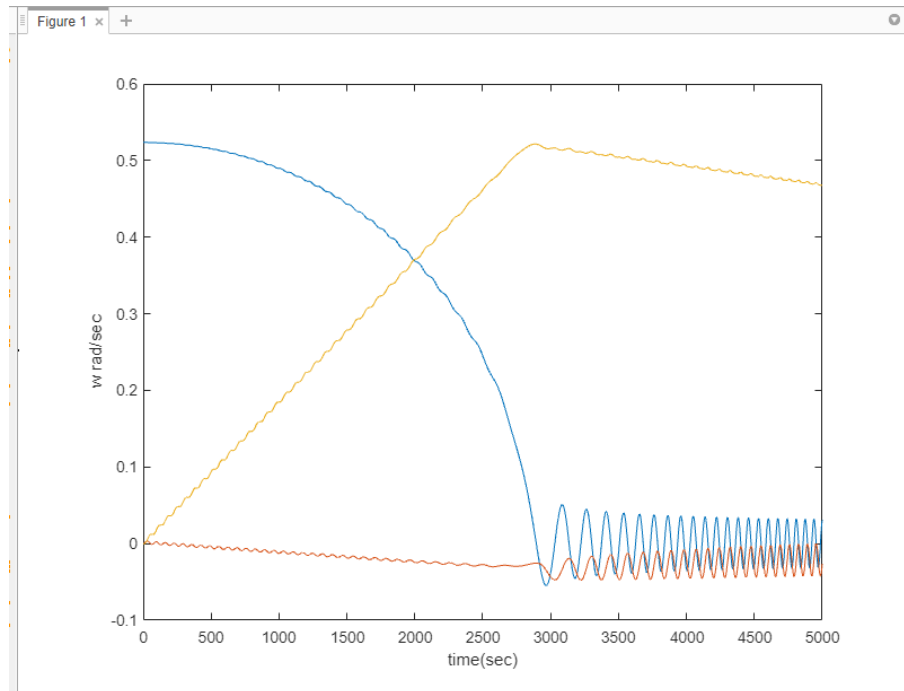


FIGURE 1 – Angular velocities

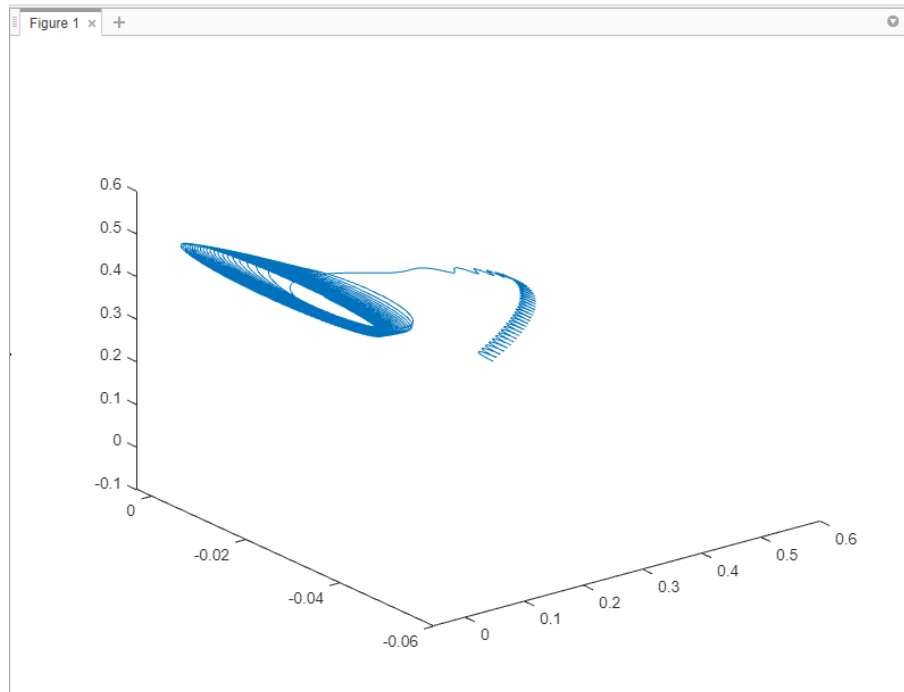


FIGURE 2 – Angular velocities 3d plot

w_1 is driven towards 0 from its initial value, while w_3 goes from 0 to the initial value of w_1 and w_2 stays close to 0 during the maneuver. The system seem to converge around this state in 3000 seconds and then starts to oscillates around these values.

The nutation angle $\beta(t)$, which is the angle between the satellite system angular momentum and the control axis also has to be plotted. To find it, Euler Parameters attitude kinematics equation is solved, then time history of Euler Parameters is converted to (3-2-1) set of Euler Angles as the nutation angle should correspond to the θ angle of Euler Angles. The initial attitude chosen for β is $\beta = [1000]^T$ and the plot obtained is the following :

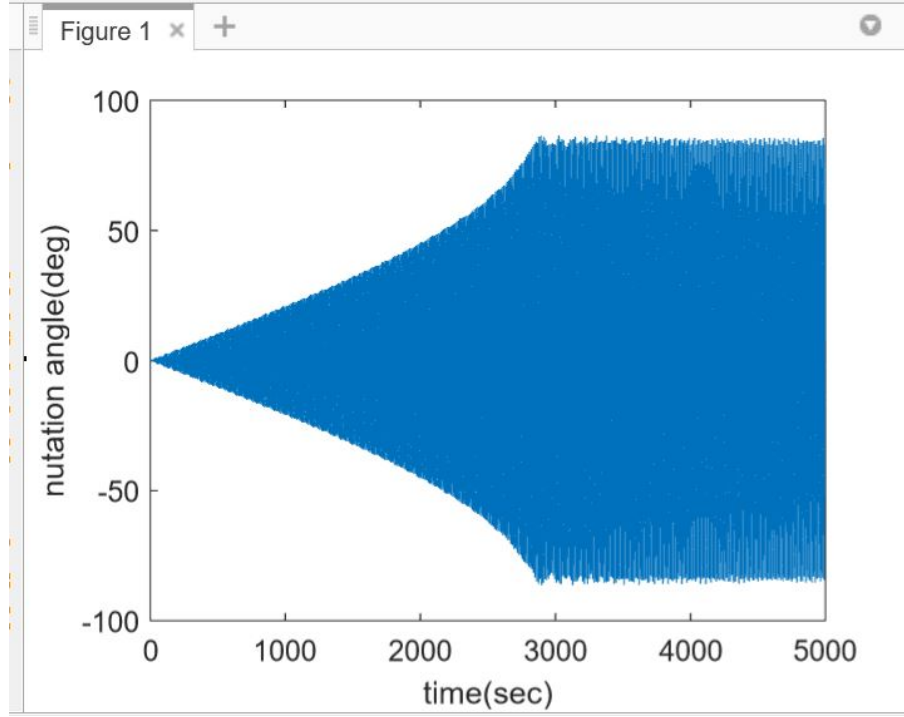


FIGURE 3 – Nutation angle

This plot is odd and shows a lot of oscillations, it might be due to an inaccurate choice of initial attitude as it should probably start at a 90 degrees angle and go towards 0 as the angular momentum vector is aligning with the control axis.

3.2 J diagonal

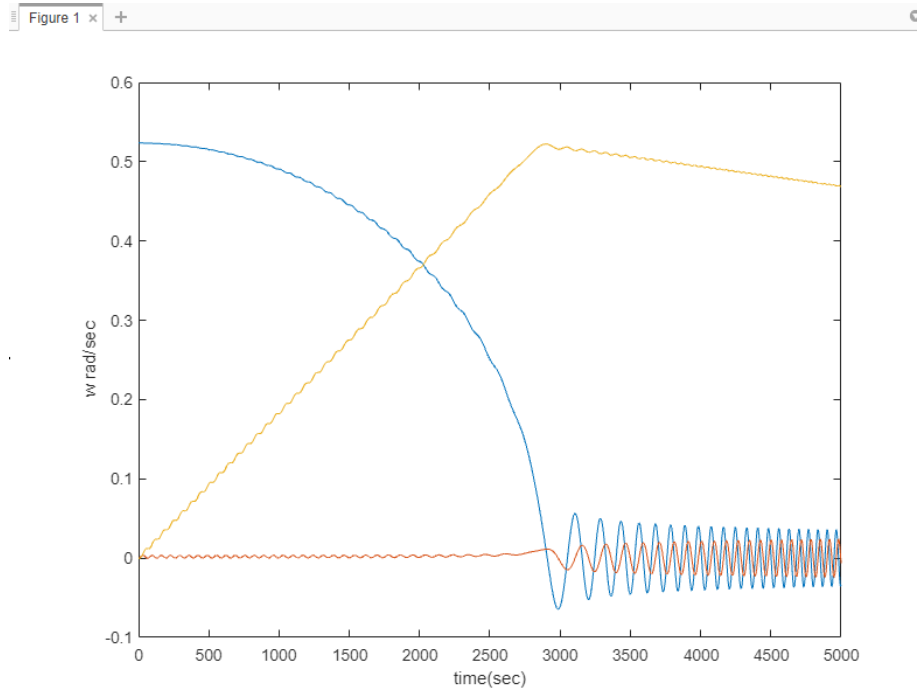


FIGURE 4 – Angular velocities

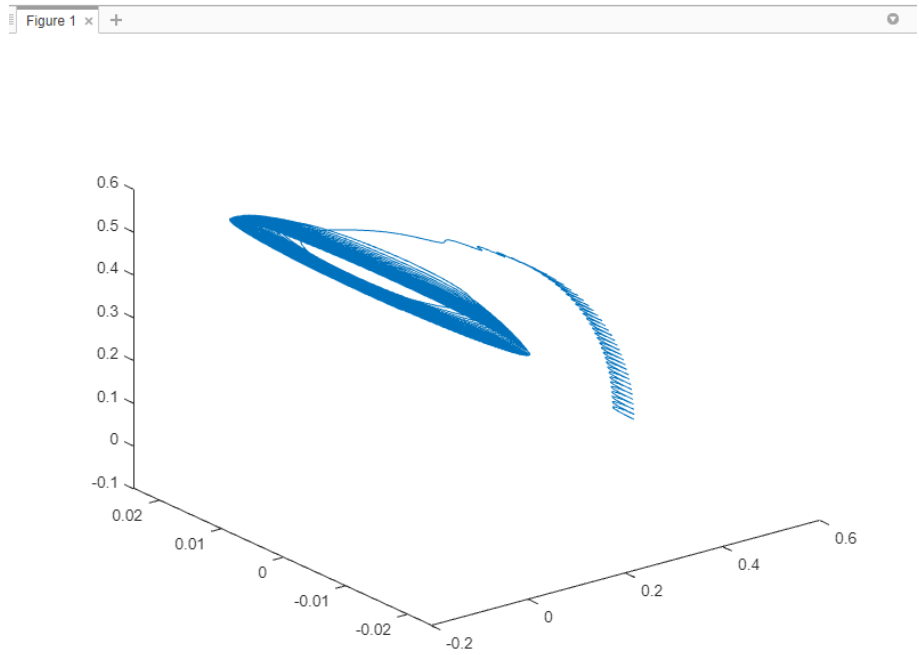


FIGURE 5 – Angular velocities

When off diagonal elements of matrix J are removed, similar plots are obtained. However, this time, the oscillations of the system after 3000 seconds are dampened, the system appears to be more stable.

4 Attitude estimation with TRIAD and QUEST

The property that $C_{BN} *^T(C_{BN}) = I$ is used.

If we calculate the resulting matrix (which is supposed to be identity in a perfect scenario) of the products of the estimated C_{BN} with TRIAD method transposed estimated C_{BN} with QUEST method, we get a resulting matrix close to I. We are supposed to get identity matrix if both method give the exact same attitude.

The attitude difference is given by the principal rotation angle corresponding to the resulting matrix. In this case, we get an attitude difference of 0.2076 degrees between QUEST and TRIAD method.

5 Stabilization

5.1 Nonlinear control law

To prove the stability, a Lyapunov function has to be found. Let's call this function V. V needs to verify $V > 0$, V continuous and differentiable and $\dot{V} \leq 0$. If we can find such function, then the stability of the dynamical system will be proved.

V needs to be function of the modified Rodrigues parameters (MRPs) and of the angular velocity :

$$V = V(\boldsymbol{\sigma}, \boldsymbol{\omega}). \quad (5.1)$$

A common idea to find candidate Lyapunov is to define it with a similar formulation to energy, as it needs to be positive. To be able to solve the equation system, QUEST solution which is expressed in classical Rodrigues parameters has to be converted to MRPs. To do this, a function is used in MATLAB. The system has to be solved for the wheel angular velocities, for the system angular velocity and also for the kinematics of the MRPs. The initial conditions are computed from the solutions of section 3-2 for the angular speeds and of section 4 for the initial position. As the goal is to stabilize the system to zero attitudes and angulars velocities, the problem is simplified and there is no need to define position and angular velocity errors here.

Let's define V such as

$$V(\boldsymbol{\omega}, \boldsymbol{\sigma}) = \frac{1}{2} \boldsymbol{\omega}^T J \boldsymbol{\omega} + 2 \boldsymbol{\sigma}^T K \boldsymbol{\sigma} \quad (5.2)$$

where K is a positive definite matrix. Therefore, if J is positive definite, it is certain that :

$$V > 0 \quad (5.3)$$

and V is continuous differentiable. Thus

$$\dot{V} = \boldsymbol{\omega}^T J \dot{\boldsymbol{\omega}} + 2 \boldsymbol{\sigma}^T K \dot{\boldsymbol{\sigma}} + 2 \dot{\boldsymbol{\sigma}}^T K \boldsymbol{\sigma} \quad (5.4)$$

Modified Rodrigues parameters kinematics equation is :

$$\dot{\boldsymbol{\sigma}} = \frac{1}{4} [(1 - \sigma^2) I_{3 \times 3} - 2[\boldsymbol{\sigma} X] + 2 \boldsymbol{\sigma} \boldsymbol{\sigma}^T] \boldsymbol{\omega} \quad (5.5)$$

Injecting this equation, it can be established that

$$\dot{V} = \boldsymbol{\omega}^T (J \dot{\boldsymbol{\omega}} + [(1 - \sigma^2) I_{3 \times 3} - 2[\boldsymbol{\sigma} X] + 2 \boldsymbol{\sigma} \boldsymbol{\sigma}^T] \boldsymbol{\sigma}) \quad (5.6)$$

To guarantee stability, V needs to verify $\dot{V} \leq 0$. To make sure of that, let's create P, the positive definite angular velocity feedback gain matrix and set :

$$\dot{V} = -\boldsymbol{\omega}^T P \boldsymbol{\omega} \quad (5.7)$$

The goal is to create a control torque to reach zero attitudes and angular velocities of the spacecraft using the reaction wheels, therefore, if the torque is called u :

$$u = -[\omega X]J_w\omega_w - J_w\dot{\omega}_w \quad (5.8)$$

From the definition of V , the following can also be concluded :

$$u = -K\sigma - P\omega \quad (5.9)$$

Therefore, to determine the reaction wheels speed to stabilize the spacecraft the following differential equation has to be solved :

$$\dot{\omega}_w = (J_w)^{-1}(K\sigma + P\omega - [\omega X]J_w\omega_w) \quad (5.10)$$

Using this V function, the stability of the system is proven and the equations system can be solved to drive the spacecraft to zero attitude and angular velocities using the defined feedback control. At this point, proper gains to have to be selected.

5.2 Gains

An idea used in the "Quaternion Feedback Regulator for Spacecraft Eigenaxis Rotations" paper was to define the gain matrix as the inertia matrix multiplied by a positive real coefficient. Therefore, matrix K and P can be defined as $K = \alpha J$ and $P = \beta J$, thus α and β will be the gain coefficients that have to be adjusted to stabilize the spacecraft.

Running the code with $K = 0.5J$ and $P = 0.6J$ provides the following results :

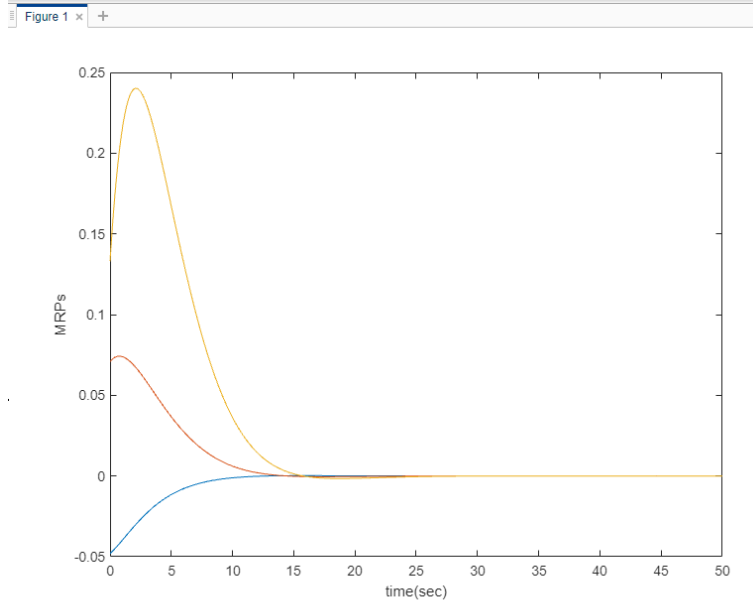


FIGURE 6 – MRPs time history

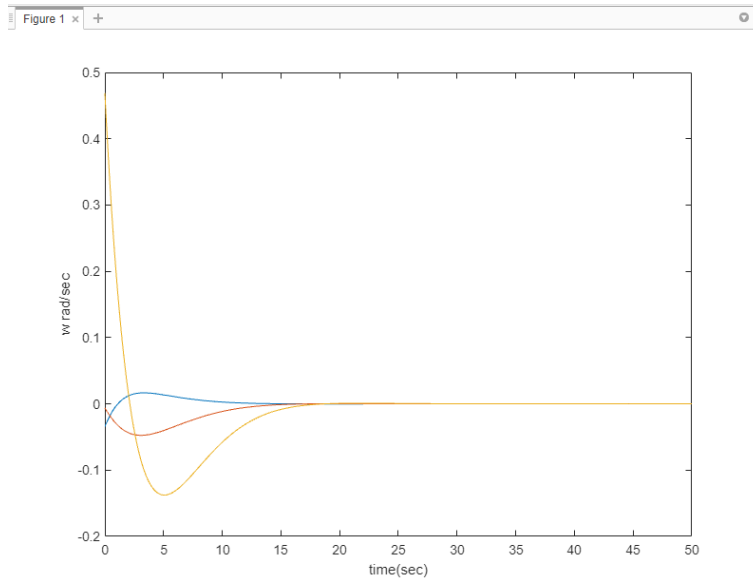


FIGURE 7 – Angular velocities time history

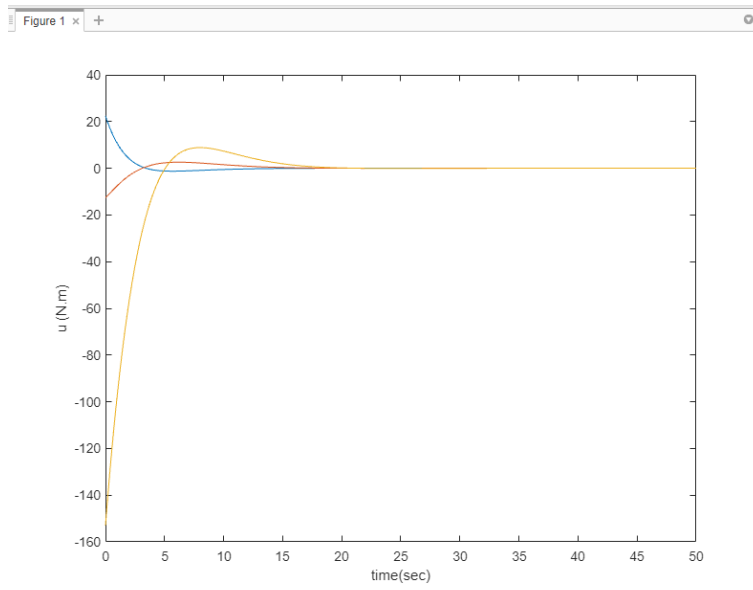


FIGURE 8 – control torque time history

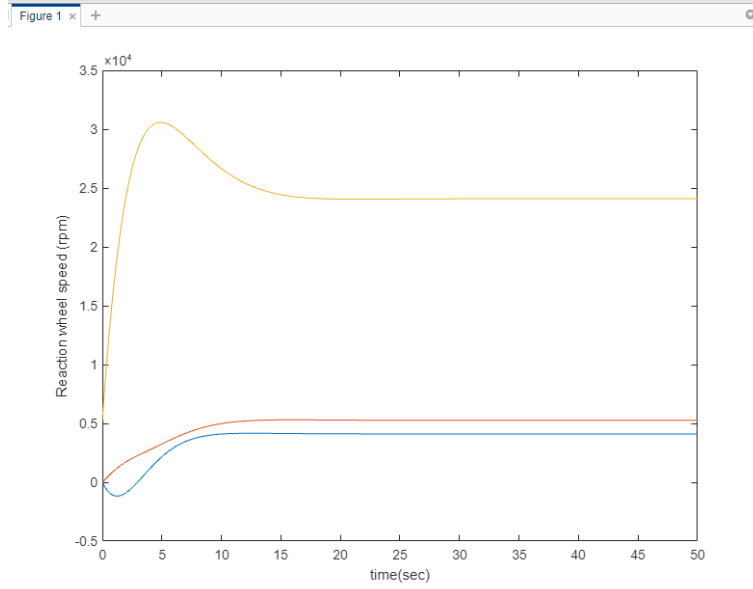


FIGURE 9 – Reaction wheels speed time history

These parameters allow the attitude and angular velocities to reach 0 in less than 50 seconds and the norm errors are less than 10^{-5} .

However, the third wheel rotation speed seem to be very high. An order of magnitude of common rotation wheel speeds in rpm is 10^3 . The first and second wheel speed are close to that value, but the third wheel speed is 10 times greater, which seems important.

This selection of gains for the feedback drive the system towards zero attitude and angular velocities, however it requires a lot of speed from the third wheel, which might be a problem for the spacecraft. Finding different gain choices or feedback control law could be a way to solve this issue.

6 Conclusion

In this project, a rigid satellite deployed in orbit and experiencing a tumbling motion was spun up along its third axis in 5000 seconds. At this moment, it had to be stabilized into zero attitudes and angular velocities in 50 seconds. To do so, MATLAB code using ode45, QUEST (and TRIAD) algorithms and a designed feedback control law have been used. Using reaction wheels to generate torque is one of the ways to control a spacecraft attitude.